UNIVERSITY OF MORATUWA
Faculty of Engineering
Department of Electronic and Telecommunication Engineering
PG Diploma/M.Sc. in Telecommunications 2019
Term 1 Examination

EN 5830 ENGINEERING DECISION THEORY

Time allowed: *Two (3)* hours February 2020

INSTRUCTIONS TO CANDIDATES:

- This paper contains 4 questions on 6 pages.
- This examination accounts for **60**% of the module assessment. The total maximum mark attainable is **60**. The marks assigned for each question & sections there of are indicated in square brackets.
- This is an **open-book** examination.
- Electronic/communication devices are not permitted. Only equipment allowed is a calculator approved and labeled by the Faculty of Engineering.
- If you have any doubt as to the interpretation of the wording of a question, make your own decision, and clearly state it.
- All questions carry equal marks.
- Answer **all** questions.

ADDITIONAL MATERIAL:

- No additional material is provided.

Question 1.

Least squares fitted equation of the following simple linear model

$$y_i = a + bx_i + \varepsilon_i, \quad i = 1, \ldots, n \tag{1}$$

is given by

$$\frac{y - \bar{\mathbf{y}}}{r s_{\mathbf{y}}} = \frac{x - \bar{\mathbf{x}}}{s_{\mathbf{x}}},$$

where $\bar{\mathbf{x}}, \bar{\mathbf{y}}, s_{\mathbf{x}}, s_{\mathbf{y}}$, and r have their usual meanings[1]. Here we assume that, ε_is are independent with $\mathbb{E}(\varepsilon_i) = 0$ and $\mathbb{E}(\varepsilon_i^2) = \sigma^2$. Moreover, we assume that x_is are non random.

(a) What are the least squares estimates of a and b (i.e., $\hat{a}$ and $\hat{b}$)? **[3 marks]**

(b) What is the most accurate statement? **[3 marks]**

i. The parameters a, b and σ^2 are observable

ii. The parameters a, b and σ^2 should be estimated from the data

(c) For a particular application, the y_is are measured in centimeters (cm) while the x_is are measured in kilograms (kg). Under this framework, what are the measuring units of $\hat{a}$ and $\hat{b}$? **[2 marks]**

(d) Assuming that the ith observation can also be written as

$$y_i = \hat{a} + \hat{b}x_i + e_i = \hat{y}_i + e_i, \quad i = 1, \ldots, n,$$

show that $\frac{1}{n}\sum_{i=1}^{n} e_i = 0$ and $\frac{1}{n}\sum_{i=1}^{n} e_i^2 = (1 - r^2)s_{\mathbf{y}}^2$. **[6 marks]**

(e) What is the most accurate statement about the e_is in (iv) for a given dataset (i.e., for known $(x_i, y_i), i = 1, \ldots, n$ pairs)? **[3 marks]**

i. The e_is are random numbers

ii. The e_is are non random numbers

iii. One cannot exactly calculate the e_is

(f) Now let us assume that $b = 0$ in (1). What is the least squares estimate of a (i.e., $\hat{a}$) corresponding to this case? **[3 marks]**

(g) Under the setting in (f), express $\frac{1}{n}\sum_{i=1}^{n} e_i^2$ as a function of $s_{\mathbf{y}}$. **[3 marks]**

(h) An unbiased estimator of σ^2 in the setting (vi) is given by $\hat{\sigma}^2 = \dfrac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1} = \dfrac{n}{n-1}s_{\mathbf{y}}^2$. Do you expect the same estimator to be an unbiased estimator of σ^2 in the more general scenario given by (1)? **[2 marks]**

i. Yes

ii. No

[1] For given vectors $\mathbf{x} = [x_1\ x_2\ \cdots\ x_n]^{\mathrm{T}}$ and $\mathbf{y} = [y_1\ y_2\ \cdots\ y_n]^{\mathrm{T}}$, we have

$$\bar{\mathbf{x}} = \frac{1}{n}\sum x_i, \quad \bar{\mathbf{y}} = \frac{1}{n}\sum y_i, \quad s_{\mathbf{x}} = \sqrt{\frac{1}{n}\sum(x_i - \bar{\mathbf{x}})^2}, \quad s_{\mathbf{y}} = \sqrt{\frac{1}{n}\sum(y_i - \bar{\mathbf{y}})^2},$$

$$r = \frac{1}{n s_{\mathbf{x}} s_{\mathbf{y}}}\sum(x_i - \bar{\mathbf{x}})(y_i - \bar{\mathbf{y}}).$$

Question 2.

Consider the equality constrained least-squares problem

$$\begin{array}{ll} \text{minimize} & ||\mathbf{A}\mathbf{x} - \mathbf{b}||_2^2 \\ \text{subject to} & \mathbf{G}\mathbf{x} = \mathbf{h}, \end{array} \tag{2}$$

where the decision variable is $\mathbf{x} \in \mathbb{R}^n$. The problem parameters $\mathbf{A}$ is a real $m \times n$ matrix, $\mathbf{G}$ is a real $p \times n$ matrix, $\mathbf{b}$ is a real m-vector, and $\mathbf{h}$ is a real p-vector.

(a) Suppose $\texttt{rank}\ \mathbf{A} = \texttt{rank}\ \mathbf{G} = n < p$. In this respect, is the problem feasible? If so what is the primal solution $\mathbf{x}^\star \in \mathbb{R}^n$ and the optimal value $p^\star$? Justify your answer. **[3 marks]**

(b) Suppose $\texttt{rank}\ \mathbf{A} = \texttt{rank}\ \mathbf{G} = n = p$. Is the problem feasible? If so what is the solution $\mathbf{x}^\star$ and the optimal value $p^\star$? Justify your answer. **[3 marks]**

(c) Suppose $\texttt{rank}\ \mathbf{A} = n$ and $\texttt{rank}\ \mathbf{G} = p$.

i. Determine the Lagrangian associated with the problem (2). **[3 marks]**

ii. Give KKT conditions for the problem. **[8 marks]**

iii. By using KKT conditions, derive expressions for the primal solution $\mathbf{x}^\star \in \mathbb{R}^n$ and the dual solution $\boldsymbol{\nu}^\star \in \mathbb{R}^p$. **[8 marks]**

Question 3.

A robot who is moving in $\mathbb{R}^2$ is to go through n gates. The ith gate, $i = 1, \cdots, n$, is characterized by the pair of points $\mathbf{a}_i \in \mathbb{R}^2$ and $\mathbf{b}_i \in \mathbb{R}^2$. Moreover, $\mathbf{a}_i$ and $\mathbf{b}_i$ are *aligned* in the direction of one axis, More specifically, $\forall i \in \{1, \ldots, n\}$, $\exists \alpha \in \mathbb{R}$ such that $(\mathbf{a}_i - \mathbf{b}_i) = \alpha[0\ 1]^{\mathrm{T}}$. For example, a scenario with $n = 7$ is depicted in Figure Q3. The width of ith gate is c_i, $i = 1, \ldots, n$, i.e., $c_i = ||\mathbf{a}_i - \mathbf{b}_i||_2$. Suppose the starting position $\mathbf{x}_0 \in \mathbb{R}^2$ and the ending position $\mathbf{x}_n \in \mathbb{R}^2$ of the robot are already specified. In particular, it is given that $\mathbf{x}_0 = (\mathbf{a}_0 + \mathbf{b}_0)/2$ and $\mathbf{x}_n = (\mathbf{a}_n + \mathbf{b}_n)/2$.

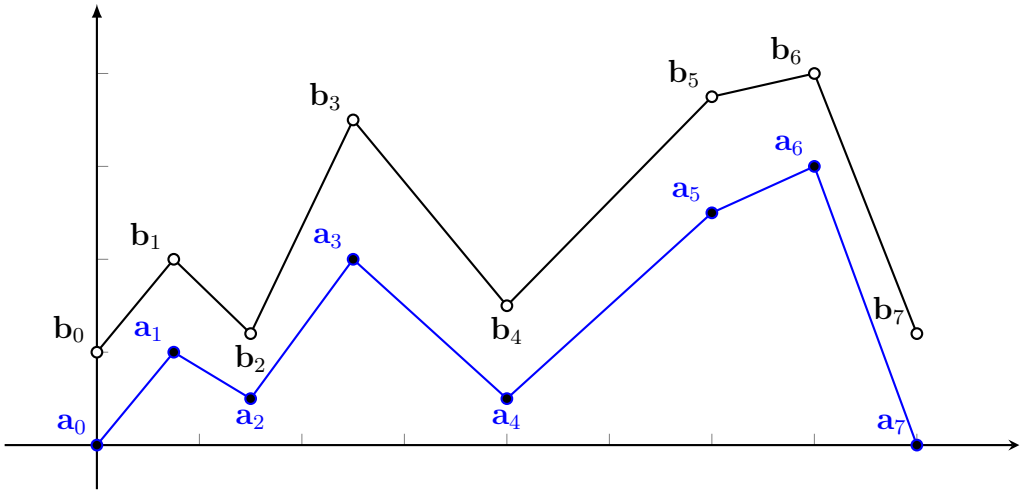


Figure Q3.: A configuration of $n = 7$ gates.

(a) Let $\mathbf{x}_i$ be a point on the line segment between $\mathbf{a}_i$ and $\mathbf{b}_i$, $i = 1, \ldots, n-1$. The goal is to find the positions $\mathbf{x}_i \in \mathbb{R}^2$, $i = 1, \ldots, n-1$, that the robot should pass through, in order to *minimize the total length* it travels from the starting point $\mathbf{x}_0$ to the ending point $\mathbf{x}_n$. Express the total length of the path of the robot, in terms of problem parameters $\mathbf{x}_0$, $\mathbf{x}_n$ and the decision variables $\mathbf{x}_i$, $i = 1, \ldots, n-1$. **[5 marks]**

[*Hint: Total length of the path can be written as a sum of lengths of some line segments.*]

(b) Clearly identify the associated constraints of the problem. **[5 marks]**

[*Hint: The condition that the robot must pass through ith gate, is characterized by using problem data* $\mathbf{a}_i$, $\mathbf{b}_i$ *and the decision variable* $\mathbf{x}_i$ *for all* $i = 1, \ldots, n-1$.]

(c) Formulate an optimization problem to determine $\mathbf{x}_1, \ldots, \mathbf{x}_{n-1}$ by clearly stating your assumptions, if any. **[5 marks]**

(d) Formulate the optimization problem in part(c) above into one of the known categories LP, QP, SOCP, or SDP. **[10 marks]**

[*Hint: You may use techniques of converting an optimization problem into its the epigraph form.*]

Question 4.

A network of 4 peer-to-peer routers labeled $n = 1, \ldots, 4$ and 8 directed links labeled $l = 1, \ldots, 8$ is shown in Figure Q4. Each router can upload or download data at a certain rate on the connection links shown in the figure. Let $x_l \in \mathbb{R}_+$, $l = 1, \ldots, 8$, be the data transmission rate of lth link in bits/s. Let $u_n \in \mathbb{R}_+$ and $d_n \in \mathbb{R}_+$, $n = 1, \ldots, 4$, denote the maximum output (upload) and the maximum input (download) data rates of nth router, respectively. Each router should conform to a *conservation of flow equation*, i.e., the sum of the input data rate equals the sum of the output data rate for all $n = 1, \ldots, 4$. Consider a scenario where there is a single data transmission, at a rate of r bits/s, from computer C_1 to computer C_2, which are connected to routers 2 and 3, respectively, see Figure Q4.

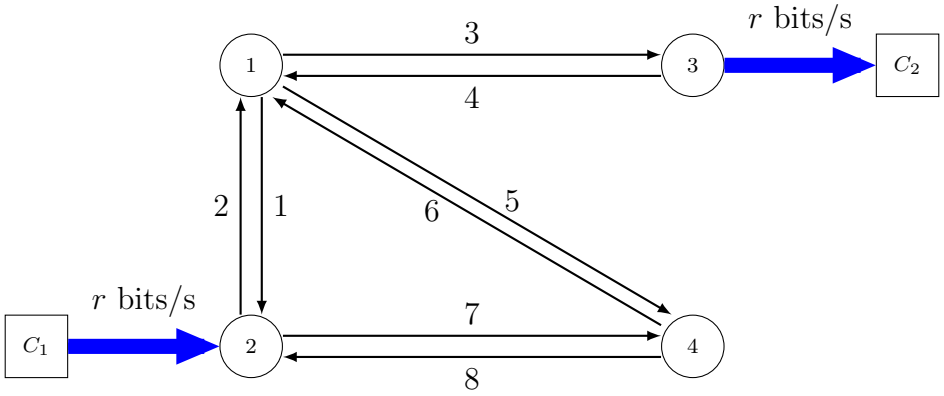


Figure Q4.: A network with 4 peer-to-peer routers.

(a) The level of congestion between two routers is defined by using the data transmission rates of the links between the two routers. In particular, the congestion between routers 1 and 3 is denoted by c_{13} and is given by

$$c_{13}(\mathbf{x}) = x_3 + x_4,$$

where $\mathbf{x} = [x_1 \ \ldots \ x_8]^{\mathrm{T}}$. The congestions between other pairs of routers are denoted, c_{12}, c_{24}, c_{14} and are similarly defined. Suppose the system designer wants to minimize the *sum of the congestions* of the network. Identify the related objective function. **[5 marks]**

(b) Clearly identify the constraints associated with the maximum data upload rate and the maximum data download rate at nth router, $n = 1, \ldots, 4$, in terms of the decision variable $\mathbf{x} \in \mathbb{R}_+^8$. **[5 marks]**

(c) Determine the flow conservation constraints at nth router, $n = 1, \ldots, 4$, in terms of the decision variable $\mathbf{x} \in \mathbb{R}_+^8$. **[5 marks]**

(d) Give a convex problem formulation to determine the data transmission rates $\mathbf{x} \in \mathbb{R}^8$ such that the sum network congestion is minimized. **[5 marks]**

(e) How will the above problem formulation change if the *worst-case congestion* of the network (i.e., the maximum congestion between the pairs of the routers) is to be minimized. **[5 marks]**

- End of Question Paper -